APAAR Verify

Platform Demonstration & End-to-End Walkthrough Guide (v1.0)

■■ DEMONSTRATION & PROTOTYPE NOTICE:
APAAR Verify is an architectural and engineering prototype designed for the Smart India Hackathon. All institutional figures, student records, and verification responses shown in this demonstration are synthetic. Identity queries are simulated and do NOT constitute an official Ministry of Education or AICTE determination.

1. Demonstration Credentials & Role Matrix

The platform enforces strict role-based access control (RBAC) with institutional data isolation:

Role	Email Username	Password	Responsibilities
AICTE Admin	admin@aicte.gov.in	Demo@1234	National monitoring, capacity ceilings, graph analytics
AICTE Reviewer	reviewer@aicte.gov.in	Demo@1234	Investigate anomalies, audit decisions
IIT Demo Operator	operator@iit-demo.edu	Demo@1234	Batch CSV uploads, draft corrections
NIT Demo Operator	operator@nit-demo.edu	Demo@1234	Independent institutional silo & uploads

2. Step-by-Step Interactive Demonstration Scenarios

Scenario 1: Institutional Batch Ingestion & 9-Rule Flagging

Context: Operator (operator@iit-demo.edu) | UI: /institution/upload

1. Log into Institution Portal as operator@iit-demo.edu.
2. Navigate to 'Upload Student Data' workspace.
3. Upload synthetic admission batch CSV; observe the 2-Stage Staging Pipeline.
4. The 9-rule integrity engine automatically flags seeded errors (APAAR_FORMAT_INVALID, REQUIRED_FIELD_MISSING, CATEGORY_INCONSISTENCY).
5. Review row-level error reports, download sanitized CSV with formula injection shielding, and commit valid records.

Expected Outcome: Batch committed with cryptographic checksum and row-level issue linkage.

Scenario 2: Multi-Stage Correction Reconciliation Workflow

Context: Operator -> Inst Admin -> AICTE Reviewer | UI: /institution/corrections & /aicte/reviews

1. From Student Detail (/institution/students/:id), Operator submits correction with proposed value & justification.
2. Inst Admin reviews institutional rationale and endorses request to AICTE review.
3. AICTE Reviewer accesses Review Queue, compares side-by-side snapshots, and confirms or adjusts decision.
4. System generates immutable StudentRecordVersion snapshot and applies update to active record.

Expected Outcome: Defensible regulatory audit trail with cryptographic versioning.

Scenario 3: Admission Integrity Graph & 30-Day Freshness SLA

Context: AICTE Admin (admin@aicte.gov.in) | UI: /aicte/graph & /aicte/integrity

1. Open Admission Integrity Graph (/aicte/graph) to visualize relationships across Institutions, Programs, and Students.
2. Inspect duplicate syndicate clusters spanning multiple institutions in real time.
3. Navigate to Source Freshness Dashboard to verify submission compliance against the 30-day regulatory SLA.

Expected Outcome: Instant visual detection of multi-enrollment rings and SLA delinquency.

Scenario 4: Asymmetric RS256 PKI Issuance & QR Verification

Context: Public Verifier / Student | UI: /verify | JWKS: /.well-known/jwks.json

1. Issue a Purpose-Bound Verifiable Credential from student profile (e.g. employment_screening).
2. Backend signs RSA-2048 token with RS256 algorithm and Key ID (kid).
3. Inspect RFC 7517 dynamic JWKS endpoint at /api/v1/.well-known/jwks.json.
4. Scan QR code or paste token into public verifier portal (/verify) for zero-trust offline mathematical verification.

Expected Outcome: Verifiable credential validation with zero raw PII disclosure.

Scenario 5: Direct Campus ERP Ingestion & Blinded ZK Proofs

Context: Campus SIS Gateway & Verifier | REST API Integration

1. Generate hashed Institutional API Key (X-Institution-API-Key: inst_live_...).
2. Push machine-to-machine records directly to /api/v1/institution-api/v1/sis/push-records.
3. Request Blinded Zero-Knowledge Proof for predicate CURRENTLY_ENROLLED_REGULAR_STUDENT.
4. Verify blinded assertion token without revealing student identity or marks.

Expected Outcome: Automated enterprise sync and zero-knowledge claim assertions.

Scenario 6: OCR-Assisted Document & Marksheet Reconciliation

Context: Operator / Reviewer | UI: /institution/students/:id (OCR Tab)

1. Open student profile and click the new 'OCR Document Reconciliation' tab.
2. Click 'Simulate Valid Marksheet' to auto-populate extracted certificate attributes.
3. Execute reconciliation to observe fuzzy string similarity and 98%+ confidence score (VERIFIED).
4. Click 'Simulate Conflicting Document' to test tampered names or conflicting roll numbers.
5. Observe automatic MISMATCH_FLAGGED status and automatic creation of a DOCUMENT_DATA_MISMATCH flag.

Expected Outcome: Automated detection of fraudulent and tampered paper documents.

Scenario 7: National Integrity Risk Studio & Anomaly Forecasting

Context: AICTE Admin (admin@aicte.gov.in) | UI: /aicte/analytics

1. Navigate to National Analytics (/aicte/analytics) and click 'Risk Scoring Studio'.
2. Inspect institutional composite risk rankings (0-100) based on anomaly density, severity weighting, duplicate clusters, and capacity excess.
3. Review Risk Tiers (Critical, High, Moderate, Low Risk) and automated regulatory audit directives.
4. Switch to 'Model Performance' tab to review precision benchmarks (96%+ average) across all 9 validation rules.

Expected Outcome: Macro-level predictive modeling and risk-prioritized regulatory review.

3. Shareable Link & Live Demo Access Instructions

To access or share the demonstration with team members and evaluators:

Service / Portal	Local Access URL	External / Shareable Access

Main Web App	http://localhost:5173	https://7f4fee2bc4e56f.lhr.life
Public Verifier	http://localhost:5173/verify	https://7f4fee2bc4e56f.lhr.life/verify
FastAPI Swagger Docs	http://localhost:8000/api/v1/docs	Interactive API testing
Dynamic JWKS Endpoint	http://localhost:8000/api/v1/.well-known/jwks.json	https://7f4fee2bc4e56f.lhr.life/api/v1/.well-known/jwks.json